Supplementary Information

Article: Input matched multisignal colocalization clarifies regulatory assignments at primary biliary cholangitis loci

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This document defines the units, provenance and interpretation limits for the S1-S11 evidence families. Supplementary Data 1 contains ten worksheet views (S1-S10); complete row-level tables are supplied in Source Data 1. The S11 external-evidence family is supplied in Source Data 2, rather than an eleventh worksheet. Worksheet and archive labels are cross-referenced below.

Supplementary upload files

- Supplementary Data 1 — Supplementary_Data_1_Tables_S1-S10.xlsx. Editable workbook view of Supplementary Tables S1–S10; the complete large row-level tables remain in Source Data 1.
- Source Data 1 — Source_Data_1_Primary_Machine_Readable_Data.zip. Full primary source-derived TSV/TSV.GZ/JSON tables and a release-level checksum manifest.
- Source Data 2 — Source_Data_2_External_Regulatory_Evidence.zip. Bounded S11 external disease-study, CASCADE, chromatin and OMIX evidence tables; no complete mediation claim.

Supplementary table legends and interpretation notes

Evidence family S1. Source and accession manifest.

Data-source manifest: accession, provider, version, availability, licence and source/derived-data hashes. It is a provenance catalogue, not an additional cohort. File map: Supplementary Data 1, worksheet S1; Source Data 1, S1/.

Evidence family S2. PBC-wide screening registry.

PBC-wide screening registry: all 6,923 locus–gene–cell comparisons, including 5,460 ABF-eligible and 1,463 coverage-insufficient rows. Rows are comparisons rather than independent genetic discoveries. File map: Supplementary Data 1, worksheet S2; Source Data 1, S2/.

Evidence family S3. Matched-input analysis of the fixed 642 comparisons.

Frozen 642-comparison A0/A1/A2/M trajectories: 184 primary triggers, 455 H3-focused tests and three borderline comparisons. Original screening labels and matched-input classifications are recorded separately. File map: Supplementary Data 1, worksheet S3; Source Data 1, S3/.

Evidence family S4. Support-set membership and alignment audit.

Source-variant membership and identity: S_A and S_M sizes, removed variants, allele alignment, ordered-variant SHA-256 and model/source identities. Equal counts do not establish equal variant identity. File map: Supplementary Data 1, worksheet S4; Source Data 1, S4/.

Evidence family S5. Source-matched LD and SuSiE diagnostics.

Source-matched QTL LD, fine-mapping and model diagnostics: PF10 primary versus PF50 sensitivity, SuSiE convergence, credible-set metrics and kriging adjudication, including retained reviewed flags. File map: Supplementary Data 1, worksheet S5; Source Data 1, S5/.

Evidence family S6. Signal-pair posterior and cross-configuration semantics.

All eligible signal-pair posteriors, prior sensitivity, cross-L identity matching and mixed H4/H3 structures. An H3-supporting pair does not establish absence of a different H4-supporting pair. File map: Supplementary Data 1, worksheet S6; Source Data 1, S6/.

Evidence family S7. Simulation and estimation channels.

Simulation parameters and outcomes: six truth-known scenarios, 486,000 matched-branch iterations, no-CS/no-pair/error channels, unconditional and pair-evaluable denominators. Single-configuration results do not directly calibrate the composite decision gate. File map: Supplementary Data 1, worksheet S7; Source Data 1, S7/.

Evidence family S8. Cross-resource molecular-QTL evidence.

Cross-resource molecular-QTL evidence, risk-allele direction and FinnGen public results. Correlated FinnGen cell strata from one resource are not independent cohort replications. File map: Supplementary Data 1, worksheet S8; Source Data 1, S8/.

Evidence family S9. Liver donor-level expression.

Donor-level HRA008003 expression and exact five-versus-five liver comparison: target–lineage CPM, permutation P values, two-target BH q values and sensitivity summaries. Biological replication unit is the donor. File map: Supplementary Data 1, worksheet S9; Source Data 1, S9/.

Evidence family S10. Claim-evidence ledger and Methods–Results mirror.

Claim–evidence ledger and the Methods–Results mirror: distinguish background, association, robust association, unavailable evidence and causal/mechanistic claim ceilings. File map: Supplementary Data 1, worksheet S10; Source Data 1, S10/.

Evidence family S11. External disease-study and chromatin evidence.

External disease and chromatin-validation module in Source Data 2: FinnGen R12/R13 disease-CS, FinnGen R13 × OneK NK/IL12RB2 sharing, fixed CASCADE coverage, partial chromatin edges and OMIX descriptive limits. Not a new gene-screen or complete mediation chain. File map: Source Data 2, data/; the constituent files retain source-stage names for provenance.

Data sharing and claim ceiling

The workbook and both Source Data archives reproduce only derived, aggregate and source-licensed shareable material. Donor genotypes, source BAMs, pseudobulk matrices and third-party licensed archives are not redistributed. Neither high posterior support nor detectable tissue expression establishes expression mediation, biological mechanism or clinical utility. All reported hypothesis and model numbers retain their frozen definitions, and the corresponding manifests identify source releases.